RansNet XE-300

4G/5G | Wi-Fi | IoT | SD-WAN

A cost-effective Industrial Wireless
Router for the Internet of Everything

RansNet XE-300 is an industrial grade wireless router designed to connect IoT/industrial devices over 3G/4G wireless network, together with RansNet Software-Defined-Branch (SD-Branch) technologies and central orchestration platform, to offer an end-to-end secure communication solution for the Internet of Everything applications.

Equipped with dedicated LTE or 5G NR modem for direct dual-SIM support, The XE-300 provides resilient and high-speed Wireless Wide-Area-Network (WWAN) backhaul as primary or backup to a fixed line, to connect remote branch devices to headquarter networks or datacenter servers.

This industrial appliance comes with multiple FastEthernet LAN & WAN ports and 802.11n Wi-Fi, with integrated GPS and GPIO connectors for IoT devices. Installed with RansNet OS, the XE-300 supports full SD-Branch features, to ensure security and end-to-end application performance, at minimum operational overheads.

The XE-300 has compact lightweight design with heatsink metal casing, easy to fit into any enclosures or mount on any machines, and is able to sustain working in harsh conditions. It is particularly useful for industrial/IoT use cases, such as backhauling video surveillance, IoT sensors, digital signage, ATM, smart locker, car parking, smart vending, EV charging, transportation, mobile hotspots.

Key features

- Dedicated LTE/5G module optimized for wireless WAN performance
- Worldwide 5G SA/RedCap, LTE, and UMTS/HSPA(+) coverage
- Integrated Wi-Fi, GPS, Bluetooth and GPIO for connecting IoT devices
- FastEthernet WAN for fixed line and LAN for local extension
- Multi-WAN for link balancing and auto failover between fixed and 4G
- Stateful firewall for perimeter defense
- Network Access Control for device security and identity management
- VPN for data encryption and transport security over the air
- Dynamic routing and traffic steering for application performance
- Built-in MQTT broker* for IoT support
- Zero-Touch-Provisioning for easy deployment
- SD-WAN orchestration for central configuration and monitoring

5G/NR (RedCap)
Max. 100 Mbps (DL)
Max. 100 Mbps (UL)

LTE CAT 3
Max. 50 Mbps (DL)
Max. 20 Mbps (UL)

802.11b/g/n
300Mbps (2.4Ghz)

FastEthernet
WAN and LAN

Multi-constellation
GNSS

MQTT Broker

Stateful Firewall

Multi-protocol VPN
(SSL/IPSec/GRE)

Software-Defined
Wide Area Network

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Hardware Specifications

Models	XE-300V	XE-300R	XE-300X
Processor	MT7628		
Memory (Flash, RAM)	32MB, 128MB		
Firewall throughput	100Mbps		
VPN throughput	10Mbps (w/ AES-256 encryption), 20Mbps (WireGuard)		
Concurrent devices	Up to 20		
Concurrent connections	Up to 5,000		
Multi-WAN	YES (Fixed line)	YES (Fixed line, 4G)	YES (Fixed line, 5G)
FastEthernet (10/100Mbps)	1 x FE WAN (PoE), 3 x FE LAN. All ports are configurable as WAN/LAN.		
Serial/GPIO	1 x RJ45 Console. 1 x RS232 + 1 x RS485 + 9 x GPIO (sold separately as an optional module)		

4G/LTE Specifications

4G/5G Modem	-	1x Quectel EM05	1x Quectel RG255
SIM Card Slot	-	Dual SIM (Active/Standby)	Dual SIM (Active/Standby)
4G/5G Bands	-	LTE-FDD: B1/B3/B5/B7/B8/B20 LTE-TDD: B38/B39/B40/B41	5G NR 3GPP Rel-17 RedCap SA operation 5G SA n1/2/3/5/7/8/12/13/14/18/20/25/26/28/30 38/40/41/48/66/70/71/77/78/79
4G/5G peak data rate (theoretical)	-	DL 50Mbps, UL 20Mbps	DL 100Mbps, UL 100Mbps
4G/5G Antenna	-	External Female SMA Connector	

Wi-Fi (2.4Ghz) Specification

Standard	802.11g/n
Wi-Fi peak data rate (theoretical)	Up to 300Mbps
Concurrent Wi-Fi devices	Up to 10 STA
Wireless Security	WPA/WPA2-PSK (TKIP/AES), WPA/WPA2 Enterprise
Bandwidth and channel	HT20/HT40, channel 1～13 (support auto)
RF Power	11n @MCS7:21±2dBm, @MCS0:24±2dBm. 11g @54M:22±2dBm, @6M:25±2dBm. 11b @11M:24±2dBm, @1M:26±2dBm.
Receive Sensitivity	11n: -70dBm@MCS7, -88dBm@MCS0. 11g: -72dBm@54Mbps, -90dBm@6Mbps. 11b: -85dBm@11Mbps, -94dBm@1Mbps.
EVM	11n: ≤-28 dB; 11g: ≤-25 dB; 11b: ≤-10 dB; 11a: ≤-25 dB
WIFI Antenna Connector	External Male SMA Connector

Physical Specification

Design form factor	Small form factor
Dimensions, Weight (WxDxH, mm)	115 x 87 x 40, < 200g
Regulatory & Compliance	CE, FCC, IP67
Power Consumption	< 12W
Operating temperature	-30°C to 70°C

Network and Security Features

IPv4, IPv6	✓
DHCP services (client and server)	✓
MQTT broker (mosquitto)	✓
VLAN/Trunking (IEEE 802.1q)	✓
Ethernet bridging (IEEE 802.1d)	
Bonding/EtherChannel/LACP (IEEE 802.3ad)	
High Availability (VRRP)	✓
Static and Dynamic routing (OSPF/BGP)	✓
Policy-Based Routing (PBR)	✓
SNMP, NetFlow, OpenFlow, Syslog, iperf3	✓
Web filtering (DNS and FQDN)	✓
Firewall, NAT/SNAT/DNAT/Port Forwarding	✓
Access logging for security audit and compliance	✓
SD-WAN Orchestration and Monitoring	✓
Cloud or on-premise SD-WAN orchestrator	✓
API for integration with external systems	✓
Zero-touch provisioning	✓
Multi-VPN: WireGuard, SSL, IPSec, GRE, VXLAN	✓
Multiple VPN topology: L2VPN, L3VPN	✓
Multiple VPN access options: star or mesh	
Ethernet over VPN (VXLAN with BGP EVPN)	✓
Auto SD-WAN VPN and routing provisioning	✓
Template based configuration and monitoring	✓
Application-based Traffic-shaping	✓
Performance-based traffic steering	✓
Network Access Control (NAC)	✓
Super-tenant multi-role based dashboard access	✓
Real-time device and SD-WAN topology update	✓
Alert on device/link failures or thresholds	✓
Extensive dashboard customization	✓
Customizable scheduled reporting	✓